



Catalysis Enhanced Pyrolysis of Blended Biomass: Effects of Blends Compositions and Catalyst Concentrations

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Abstract

The oleic industry is one of the most polluting anthropogenic activities in the Mediterranean basin generating huge amounts of solid biomass residues. This study investigates the impact of pine sawdust addition and potassium catalyst concentrations on the pyrolysis behavior of olive mill solid wastes composed of olive pomace and seeds. Four types of lignocellulosic biomasses were prepared from exhausted olive mill solid residues (G) and pine sawdust (S) in different proportions: 100% G (100G), 60%S+40%G (60 S.40G), 50%S+50%G (50 S.50G) and 80%S+20%G (80 S.20G)). Three concentrations of K_2CO_3 were considered during impregnation phase: 0.1 M, 0.5 M and 1.5 M respectively. A significant catalytic influence of potassium was observed up to 0.1 M for samples with a high percentage of pine sawdust (60 S.40G and 80 S.20G). However, for the other samples (100G and 50 S.50G) the catalytic effect of potassium was observed up to 0.5 M. Moreover, the char yield was considerably enhanced by potassium impregnation, which doubles for the blended biomasses (rising from approximately 21% to 45%) at the highest concentration of 1.5 M. To better interpret these observations, the surface properties of the char were examined using Scanning Electron Microscopy with Energy Dispersive X-ray spectroscopy (SEM-EDX) and Fourier Transform InfraRed spectroscopy (FTIR). SEM analysis showed that the char generated from pure olive mill residues (100G) exhibited severe surface melting and structural sintering at a highly potassium impregnation concentration (1.5 M). Indeed, beyond a certain threshold of potassium concentration (0.5 M), agglomerations occurred blocking the active sites of the chars.

Keywords Blended biomass · Impregnation · Pyrolysis · Catalysis

1 Introduction

Lignocellulosic biomass is a major renewable and economically viable resource to meet the rising global energy demand while reducing the environmental impacts of fossil fuels [1, 2]. In the Mediterranean basin, the olive oil industry represents a key economic sector, but it is also among the

most polluting human activities due to the massive generation of agricultural byproducts [3]. In Tunisia, for instance, which ranks as the fourth worldwide producer of olive oil, this industry generates approximately 400.000 tons of olive solid waste each year [3]. Thermochemical conversion processes, notably pyrolysis, provide a highly effective strategy to upgrade these residual solids into useful gaseous, liquid and solid products [4–6]. The selectivity and properties of these end-products depend essentially on the operating conditions such as temperature, heating rate, biomass conditioning, and on inherent compositional fractions of the biomass; cellulose (40–50%), hemicellulose (15–20%) and Lignin (20–30%) respectively.

Biomass matrices also contains low concentrations of minerals primarily alkali and alkaline earth metals such as potassium (K), calcium (Ca), sodium (Na), magnesium (Mg), which endogenously influence thermal decomposition [6]. Previous studies have proven that these metals exhibit strong catalytic activity during pyrolysis. For

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instance, Wan et al. [7] pointed out that alkali and alkaline earth metals significantly decrease characteristic pyrolysis temperatures and activation energies. Specifically, the form and concentration in which potassium is added play a critical role. Indeed, as demonstrated by Guo et al. [8], even low amounts of potassium loaded on pine wood can enhance H_2 and CO_2 production. Moreover, the potassium addition to wheat straw has shown the hemicellulose and cellulose degradation peaks to lower temperatures while increasing the char yields by approximately 20 wt% [9]. On the other hand, the mobility and distribution of potassium in the biomass's matrix are highly temperature dependent. In deed, as reported by Knudsen et al. [10], when the straw is subjected to pyrolysis temperatures above 700 °C, a significant fraction of potassium is released from its original binding sites. It can either volatilize or become incorporated into resulting char matrix as KCl, K_2CO_3 or potassium silicates.

Furthermore, the presence of inorganic additives deeply affects the structure of the biochar. Research on model components such as alkaline lignin or palm shells reveals that chemical impregnation with potassium compounds (such as KOH) induces a particular morphological structure. This matrix features several cavities and pores of varying sizes and shapes, which are formed by escape of volatile gases [11, 12]. Such engineered, porous biochars are highly valuable for soil amendment and carbon sequestration [13, 14].

However, while existing literature provides extensive insights into individual feedstocks like straw, pine, rice husk. or specific model molecules [9, 15], a pronounced research gap persists regarding the interactive and synergistic effects of potassium impregnation on multi-biomass blends. The structure and morphological evolution of chars derived from pyrolysis mixtures under varying catalyst concentrations remains largely unexplored.

In the present study, a series of experiments were conducted to extensively analyze the catalytic effects of potassium on biomass pyrolysis. The process of chemical impregnation was used to prepare biomass rich of potassium with various concentrations of K_2CO_3 . In addition, potassium's impact on the structure of chars was assessed through; scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM-EDX) and Fourier-transform infrared spectroscopy, especially as such information are relatively rare in literature. To achieve this aim, three different concentrations of aqueous solution of K_2CO_3 were chosen for the impregnation of exhaustive olive mill solid waste, either mixed with or without pine sawdust in different percentages. From a scientific perspective, the results we hope to achieve could help in apprehending how biomass broke down when potassium acted as a catalyst, and

could offer valuable information for improving the reactors design and facilitate operating conditions. From an application perspective, the results are intended to foster biochar production with the aim of applying it as soil amendment, both improving agricultural yields and sequestering carbon dioxide.

2 Materiel and Methods

2.1 Sample Preparation

The biomass chosen for this research consists of olive mill solid waste (abbr. G for grignon) gathered from the Zouila Oil Press Company located in the Mahdia area (Sahel of Tunisia). Additionally, pine sawdust (abbr. S) was supplied by a woodworking factory installed in Mulhouse region (France). Mixtures were made following the same method that was previously described [16]. Four types of biomasses were prepared as pellets.

- 100G constituted by pure olive solid waste (100%).
- 50 S.50G composed of 50% pine sawdust and 50% olive solid waste.
- 60 S.40G made up of 60% pine and 40% olive solid waste.
- 80 S.20G composed of 80% pine and 20% olive solid waste.

To achieve wet impregnation of potassium into biomass, aqueous K_2CO_3 solution was used. Indeed, according to reported studies, among the different salts used for impregnation, carbonate was the one that had the least influence on the pyrolysis process [17]. Therefore, the significant effect observed during the tests could be related to potassium ion K^+ and not to CO_3^{2-} . This fact is in strong agreement with the fundamental mechanisms reported by Radovic [18]. After dissolving potassium carbonate salt in deionized water to prepare three concentrations; 0.1 M, 0.5 M and 1.5 M respectively, the four samples were submerged in these solutions. Based on literature, this specific range was chosen to investigate the effect of low (0.1 M), intermediate (0.5 M) and high (1.5 M) concentrations on the pyrolysis process of the blends. The pH of the impregnation solutions was not adjusted in order to avoid the introduction of foreign chemical counter-ions into the biomass matrix. These mixtures were agitated for 12 h at room temperature ($21 \pm 1^\circ C$) using an overhead stirrer set to 500 rpm. The room temerature was chosen on the basis that heating large volumes at large industrial scale-up would lead to massive energy consumption and high operational coasts. Then, a long contact time was

implemented ensuring efficient diffusion and distribution of potassium throughout the biomass matrix. To prevent a sample layer from forming at the bottom of the beaker, the solution to biomass ratio was kept at 16 ml/g and which was selected to ensure a complete wetting and homogenous impregnation of the texturally diverse biomass matrices. Once the impregnation process was complete, vacuum filtration was used to obtain the final impregnated samples. Finally, the samples were dried for 12 h at 105 °C. To ensure the reproducibility during impregnation, a single large volume stock solution for each specific concentration (0.1 M; 0.5 M and 1.5 M) was prepared. The biomass feedstock was then divided into multiple identical fractions and the exact same impregnation procedure was repeated three times using batches from these master solutions. This approach systematically minimized experimental and human errors.

Samples will be named as follows. Each sample name (i.e. 50G.50 S) followed by the impregnation level (N.I. for not impregnated and 0.1 M, 0.5 M and 1.5 M for the different concentrations of K_2CO_3 used in the impregnation solutions). It is worthy to justify the choice of K_2CO_3 among other potassium salts such as KOH or KCl for examples. Indeed, KOH is a strong base which would alter and even dissolve the organic matrix. On the other hand, KCl was because during pyrolysis, the chloride anions could volatilize into hazardous compounds and hydrochloric acid gas.

The outcomes of analyses conducted using Inductively Coupled Plasma-Optical Emission Spectroscopy (ICP-OES) on the samples prepared to measure mineral quantities are shown in Table 1. It is to be highlighted that the mineral analysis for the 100G and 100 S (100% pin saw-dust) samples was previously performed [19].

The ultimate analysis of biomasses was done using an elemental analyzer to find the weight percentages of C, H, and N in the biomass. The weight% of oxygen is deduced by difference as: $O (wt\%) = 100 - (\%C + \%H + \%N)$. Results are listed in Table 2.

Results of proximate analysis of non-impregnated samples, which were determined in a previous study [16], are summarized in Table 3.

The distribution of mineral particles on the surface of the chars (obtained by pyrolysis, described hereinafter) is analyzed by Scanning Electron Microscopy (SEM) and the chemical composition of different char's surfaces was performed by Energy Dispersive X-ray (EDX). The results are summarized in Table 4.

The hemicellulose content in olive mill solid waste varies between 21.5% and 23.6% of the total weight. Cellulose content is between 24% and 24.6% and the lignin fraction varies between 38% and 48.4% [20, 21].

Table 1 Inorganic elements analysis of different types of biomasses (mg/kg of dry matter)

SpL: Conc.	100G	100 S	50 S.50G			60 S.40G			80 S.20G		
	N.I	N.I	N.I	0.1 M	0.5 M	1.5 M	N.I	0.1 M	0.5 M	1.5 M	N.I
Na	3637	10	831	951	3275	424	803	747	358	672	524
K	17,189	1406	<9804	19,938	38,893	118,968	<10,204	22,111	40,837	84,691	<6250
P	615	64	564	831	360	334	566	632	318	546	417
Mn	8	48	<49	<45	<18	<17	25.5	<34	21.6	<29	25.7
Fe	197	49	372	247	265	205	447	564	198	208	172
Mg	551	113	330	236	167	106	340	225	156	147	312
Si	416	45	2987	3104	43,754	2064	1691	2094	1955	2670	928
Ca	5301	1397	3612	3391	2803	2229	3687	3138	3038	2651	2231
Al	226	34	165	146	1574	213	132	368	169	423	106
N.I.: non-impregnated											860
											95.3
											742
											84,945
											621
											<30
											297
											103
											762
											1496

Table 2 Ultimate analysis of different biomasses

Samples		N, %	C, %	H, %	O, %
100 S	NI	0.1	47.17	6.10	46.27
100G	NI	1.16	46.43	6.38	46.03
	0.1 M	1.21	49.90	6.58	42.31
	0.5 M	0.74	41.54	5.79	51.93
	1.5 M	0.94	29.83	4.05	65.18
80 S.20G	NI	0.21	45.56	6.38	47.64
	0.1 M	0.21	43.98	6.21	46.68
	0.5 M	0.20	41.15	5.88	43.76
	1.5 M	0.16	37.11	5.10	40.20
60 S.40G	NI	0.30	46.20	6.42	43.84
	0.1 M	0.29	44.30	6.03	42.70
	0.5 M	0.27	42.02	6.06	42.90
	1.5 M	0.23	37.81	5.20	39.86
50 S.50G	NI	0.35	46.27	6.51	45.85
	0.1 M	0.34	45.38	6.42	44.16
	0.5 M	0.29	39.56	5.55	39.89
	1.5 M	0.23	34.09	4.74	45.71

Table 3 Proximate analysis of non-impregnated samples

Samples	%M	%VM	%FC	%Ash
100 S	7.90	77.60	14.19	0.30
100G	38.34	45.60	10.86	5.20
80 S.20G	10.60	71.60	16.80	0.98
60 S.40G	9.30	74.70	14.11	1.89
50 S.50G	11.30	70.50	16.23	1.97

M is the moisture, VM is the volatile matter, FC is the fixed Carbon

2.2 Experimental Device

The samples were subjected to slow pyrolysis in a Macro-Thermo-Gravimetric reactor, as illustrated on Fig. 1 and described in detail previously [22]. The experimental device consists of a vertical electrically-heated alumina cylindrical reactor with a length of 2 m and an internal diameter of 0.75 m, divided into three independently controlled heating zones to ensure temperature homogeneity. It has been designed to work on real particles. It can work with particles up to 5.5 cm size. For each test, approximately 400 mg of sample was loaded. The temperature of pyrolysis process was 400 °C with a heating rate of 10 °C/min. The choice of slow pyrolysis and the specific final temperature aimed to achieve the highest possible char yield. Before removing char from the reactor, the sample should be cooled to room temperature under a flow of N₂ to avoid self-ignition under air. The gas flow rate was set at 7.8 NL/min.

For pyrolysis experiment, the conversion level (X) at a given time t is calculated according to the following:

$$X(t) = \frac{m_i - m_t}{m_i - m_f} \quad (1)$$

Table 4 Summary of the EDX results on chars

Element (%Mass)	100G				50 S.50G				80 S.20G			
	NI	0.1 M	0.5 M	1.5 M	NI	0.1 M	0.5 M	1.5 M	NI	0.1 M	0.5 M	1.5 M
C	68.53	74.54	71.71	58.09	80.44	78.30	61.63	68.63	76.46	72.74	55.66	46.78
O	25.39	21.33	21.99	25.96	17.18	19.55	23.43	21.89	22.11	20.54	25.85	30.02
Al	10.50	0.08	0.06	0.06	0.10	-	-	-	-	-	-	-
Si	0.14	0.38	0.09	0.07	0.28	-	-	-	-	-	-	-
P	0.11	0.17	-	-	0.06	-	-	-	-	-	-	-
K	4.59	1.19	5.14	14.15	0.79	1.86	14.94	9.48	0.75	6.72	16.88	23.21
Ca	0.74	0.74	0.96	1.62	0.95	0.29	-	-	0.69	-	1.61	-
Na	-	0.26	-	-	0.12	-	-	-	-	-	-	-
Mg	-	0.21	0.06	0.06	0.09	-	-	-	-	-	-	-
S	-	0.11	-	-	-	-	-	-	-	-	-	-
Total	100	100	100	100	100	100	100	100	100	100	100	100

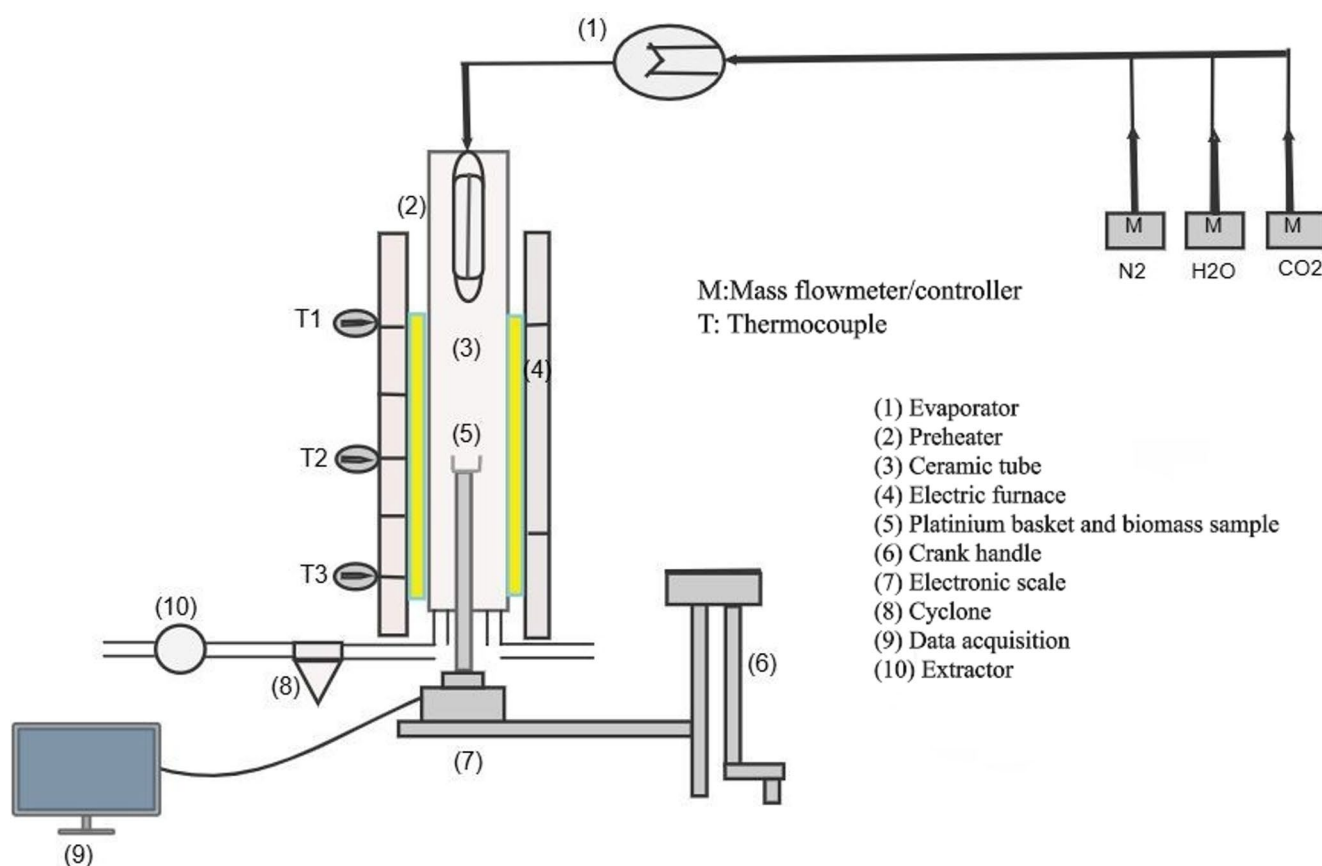


Fig. 1 Representation of the Macro Thermo-Gravi-Metric (M-TG) reactor

Where m_i represents the initial mass, $m_t = m(t)$ represents the sample's mass measured at time t and m_f corresponds to the final mass.

The pyrolysis rate is defined as the variation of the conversion versus time:

$$r(t) = \frac{dX}{dt} = \frac{X_{t+1} - X_t}{\Delta t} \quad (2)$$

Where Δt is the time difference between two successive measurements, which was set to 0.2s during data acquisition.

To investigate the surface properties, the metal distribution and the structural characterization of the chars, we used Scanning Electron Microscopy (SEM) coupled with energy dispersive X-ray spectroscopy (EDX). SEM analysis was performed using a Quanta 200 FEG microscope, operating under high vacuum at an accelerating voltage of 15 kV.

To qualitatively investigate the effect of potassium impregnation on the samples chemical structures, the chars were analyzed using infrared (IR) spectra based on the method described by Zhao et al. [23] recorded at the absorbance mode between 4000 and 400 cm^{-1} (in the mid infrared range) with a Fourier transform infrared spectrophotometer (BX FTIR system spectrometer, PerkinElmer,

MA, USA). All the spectra were obtained at a resolution of 2 cm^{-1} .

3 Results and Discussions

3.1 Biomass Characterization

To analyse the thermal response during pyrolysis, a physico-chemical characterization of the raw and impregnated biomass samples was conducted. Table 1 presents the inorganic elemental compositions determined via Inductively Coupled Plasma-Optical Emission Spectrometry (ICP-OES). The results reveal an important increase in potassium content providing clear evidence of successful impregnation.

Tables 2 and 3 outline the ultimate and proximate analysis results of the 4 prepared samples. Ultimate analyses shows a low nitrogen content and standard contents of C, H and O for lignocellulosic biomasses. Proximate analyses exhibit important %VM and %FC values, which will affect the char yield during pyrolysis, and a notable decrease of ash content for samples resulting from olive mill solid waste and pine sawdust blends. This is due to the the small ash content in pine sawdust.

Table 4 summarizes the Energy-Dispersive X-ray Spectroscopy (EDX) useful for analysing the quantitative elemental distribution and mineral retention on the surface of the resulting chars. The analysis confirms a strong surface enrichment of potassium peaking in the 1.5 M samples.

Based on the comprehensive data synthesized in these tables, a detailed analysis of biomass components and the specific chemical transformations induced by potassium is discussed below.

3.2 Char Properties and Yields

Proportions of pyrolysis products depend on operating conditions such as temperature, heating rate, residence time, as well as on the physicochemical properties of the biomass [24–26]. In pyrolysis experiments, the DTG curve of olive mill wastes exhibited two peaks with almost the same intensity: the first one appearing approximately at $T=262\text{ }^{\circ}\text{C}$, corresponding to the degradation of hemicellulose, and the second one appearing at $T=332\text{ }^{\circ}\text{C}$, corresponding at cellulose degradation. For the pine sawdust, the two peaks corresponding to the degradation of hemicellulose and cellulose appeared at temperatures of $296\text{ }^{\circ}\text{C}$ and $346\text{ }^{\circ}\text{C}$, respectively. For lignin, degradation begins from $190\text{ }^{\circ}\text{C}$ and continues up to $900\text{ }^{\circ}\text{C}$. Therefore, at a selected pyrolysis temperature equal to $400\text{ }^{\circ}\text{C}$, the degradation of all the biomass used was almost complete, except for lignin, as complete lignin degradation takes place at higher temperatures.

Increasing the pyrolysis temperature beyond a certain threshold value has a negative impact on char yield. This behavior is supported by the findings of Jindo et al. [27], who reported that increasing the pyrolysis temperature from $400\text{ }^{\circ}\text{C}$ to $700\text{ }^{\circ}\text{C}$ resulted in a decrease of char production by 16% for woody material and by 15% for rice straw. Similarly, Kan et al. [28] showed a reduction in wood char production from 39% to 27% by increasing the pyrolysis temperature from $400\text{ }^{\circ}\text{C}$ to $800\text{ }^{\circ}\text{C}$. Therefore, in order to increase the char yield obtained during the pyrolysis process, a pyrolysis temperature of $400\text{ }^{\circ}\text{C}$ should be adopted.

Most of the mineral substances in biomass are found as inorganic species, such as K, Na and Ca, which are bound to organic species or are found as salts within the anatomical structure of the biomass [29]. Many authors have studied the effects of these inorganic substances, either initially present in the biomass or added by impregnation [30–32]. These minerals impact the mechanisms involved and the yields of pyrolysis products, in particular char yields and their physicochemical properties [33]. Among the metals present in biomass, potassium is the inorganic species that appears to have the greatest influence on thermal degradation mechanisms [19, 34].

According to Table 5, potassium impregnation increases the char yield of the different samples. In fact, from the raw

Table 5 Char yields (%) of different biomasses

[K ₂ CO ₃] Samples	N.I	0.1 M	0.5 M	1.5 M
80 S.20G	21.18	28.05	34.36	41.55
60 S.40G	22.53	27	32.41	45.22
50 S.50G	21.75	27.93	36.38	45.19
100G	28.84	33.49	38.92	39
Mean $\pm$ SD	23.58 ± 3.55	29.12 ± 2.95	35.52 ± 2.78	42.4 ± 3.01

“Mean $\pm$ SD line reflects the overall statistical average and variability across all biomass formulations at each concentration level”

sample to an impregnation of 1.5 M of K₂CO₃, the percentage of char corresponding to 80S20G, 50S50G and 60S40G was almost doubled. However, for the sample 100G, the char yield increased by 10% relative to its baseline, rising from an initial value of 28.84% to 39% upon potassium impregnation. It is to be noted that 100G contains the highest amount of naturally existing potassium, while 100 S contains the lowest amount of potassium. Also, according to the results of Tables 1 and 4, after impregnation, a significant increase in potassium content was observed in all biomasses. In addition to potassium, variations were also detected in the concentrations of elements such as calcium, magnesium, sodium and silicon. From a quantitative perspective, the increasing char formation shows a good proportionality with the potassium content ($R^2=0.895$, Fig. 2), with the char yield increasing by $0.225 \pm 0.039\%$ per mg/kg of potassium in the initial biomass content.

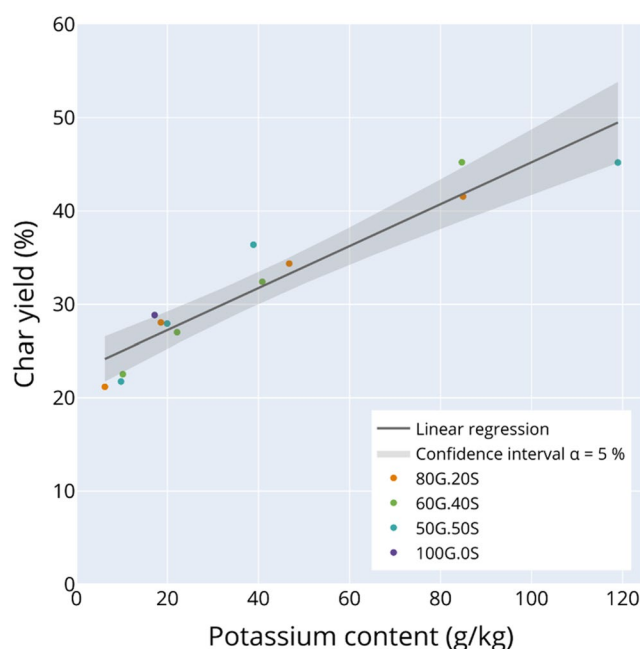


Fig. 2 Char yield versus potassium content. Linear regression obtained using Ordinary Least Square. Fitting line equation: $y = a + bx$, with $a = 0.225 \pm 0.039\%$ char/(mgK⁺/kg_{Biomass}), $b = 22.75\%$ char, $R^2 = 0.895$, $p\text{-value} = 1.0 \cdot 10^{-6}$

The observed increase in solid yield could be linked to the fact that potassium aids the opening of rings and the breaking of glycosidic bonds. It is highly probable that potassium enhanced the char production by changing the paths of pyrolytic reactions and by promoting polycondensation reactions. In addition, it has been proven that potassium impregnation facilitates the polycondensation reactions which are responsible for char formation, and accelerates the dehydration of the samples. Consequently, the recombination of volatile species gives an important char yield [27]. This acknowledges that the final char weight is a result of both the organic matter produced via polycondensation from retained minerals including potassium.

Regarding the specific behavior of the samples, the small increase in char formation for sample 100G after potassium addition compared to the other samples is due to the fact that 100G NI already contains a significant amount of potassium. Moreover, the activity of inorganic elements is a function of the number of active sites available in the biomass, and the reactions that lead to char formation are initiated from these sites. So, probably, the active sites of the 100G sample are not all available to the impregnated potassium since this sample is already rich in potassium. Furthermore, the variations detected in the concentrations of other elements (Ca, Mg, Na, Si) could be attributed to leaching effects during the impregnation process or to ion exchange mechanisms occurring within the organic framework.

This promotional effect of potassium on char yield is in agreement with Hao et al. [15] and Yang et al. [28], who observed that the addition of potassium to hemicelluloses

significantly promotes char formation, although they did not specify a threshold value. Also, Di Blasi [35] reported that significant amounts of minerals in the biomass engender an increase in solid and gas yield during pyrolysis. Similarly, Nassar [36] has shown that the addition of alkaline salts, including potassium, to bagasse and rice straw promotes the formation of char. Conversely, Nowakowski and Jones [37] observed no effect of potassium addition on the char yield of xylan during pyrolysis, highlighting that the exact matrix composition can influence catalytic pathways.

3.2.1 Catalysis Effect of Potassium

In previous studies, two approaches are used for sample preparation before impregnation: washing the samples before impregnation [38] or not [39]. From a practical point of view, washing the sample might change the biomass structure and make it harder to use on a larger scale. Consequently, the biomass was directly impregnated without any prior washing step.

The experimental results from the pyrolysis of 80 S.20G sample are displayed in Fig. 3. Figure 3a shows the conversion graph as described by Eq. 1 and Fig. 3b presents the conversion rate computed according to Eq. 2. Before any impregnation, the sample (80 S.20G) exhibited the lowest initial potassium amount among all mixtures, second to the raw pine sawdust baseline (100 S) as previously detailed in Fig. 2.

In Fig. 3b, the peak of the curve represents the maximum rate of biomass degradation. It can be seen that the

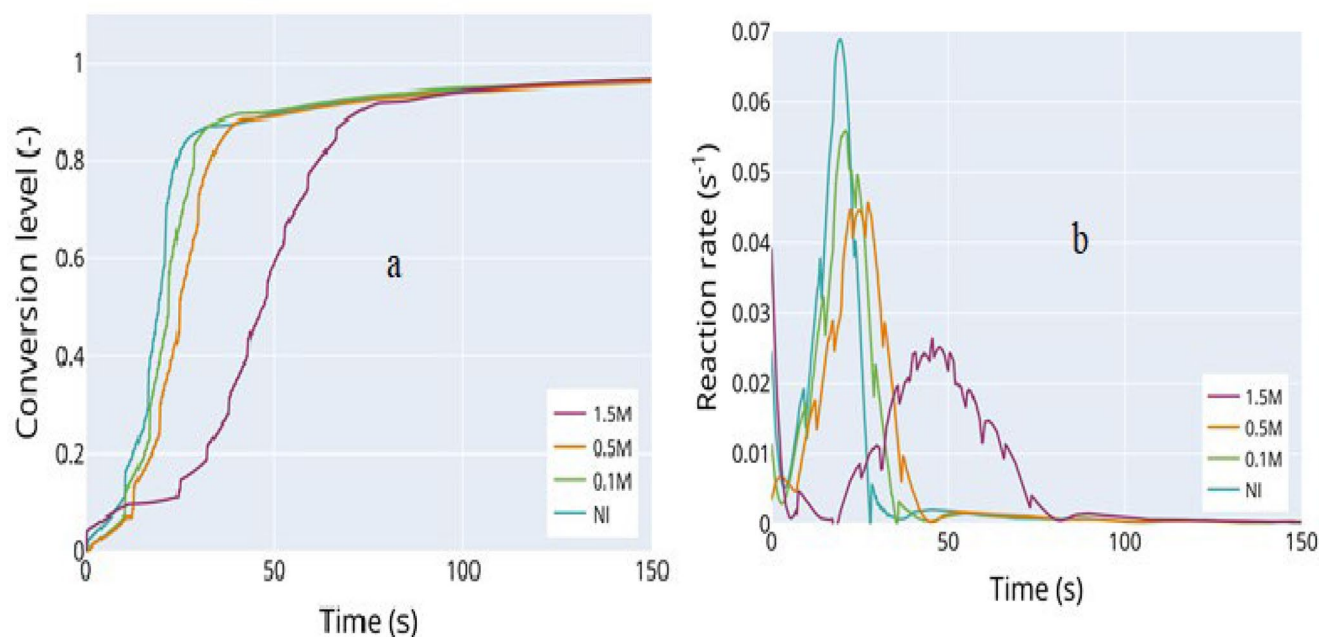


Fig. 3 Variation of the conversion (a) and the conversion rate (b) of 80 S.20G biomass as a function of time with different potassium concentrations during pyrolysis

rate of degradation rises as the concentration of impregnated potassium increases. Specifically, the peak of the pyrolysis curve of the non-impregnated sample is situated at $t_1 = 45.58$ s with a conversion rate equal to 0.07 s^{-1} and it shifted toward 0.05 s^{-1} at $t_2 = 22.33$ s for 0.5 M potassium impregnation (the half-time is decreased by half). Thus, to reach 80% conversion, unimpregnated biomass takes 62 s, while the one impregnated with potassium at a concentration of 0.5 M takes only 29 s.

Figure 4 shows the behavior during pyrolysis of the other samples with and without impregnation with different concentrations of K_2CO_3 solutions. It can be clearly observed that the pyrolysis curves of the two samples (80 S.20G and 60 S.40G) approximately exhibit similar behaviors. In fact, potassium impregnation, with a concentration equal to 0.1 M, had a significant effect on decreasing biomass degradation time. On the other hand, increasing impregnation concentration in these biomasses had little impact, as the conversion curves for the three impregnated samples (0.1 M, 0.5 M and 1.5 M) were almost overlapping.

Moreover, as we increase the fraction of olive pomace in the biomass studied (50 S.50G, 100G), the effect of potassium decreases considerably: there is a slight difference between curves before and after impregnation and this up to a concentration equal to 0.5 M. If we increase the potassium impregnation further to 1.5 M, we notice that the conversion curve is superimposed on that of 0.5 M for the 50 S.50G biomass. Whereas for the 100G biomass, potassium impregnation with a concentration of 0.5 M slightly reduced the pyrolysis time, but the increase in potassium addition to 1.5 M shifted the curve to the right.

Curves in Fig. 5 representing the conversion rate of 100G and 50 S.50G illustrate this effect well. Indeed, the conversion rate for 100G biomass reveals a sharper peak when the impregnation concentration is set at 0.5 M; however, when the potassium concentration rises to 1.5 M, the degradation peak becomes less pronounced and overlaps with the peak associated with the degradation of the biomass that has not been impregnated. Then, for 50 S.50G biomass, the two peaks relating to biomass degradation for $C = 0.5$ M and for $C = 1.5$ M coincide. This comparison is further observable in

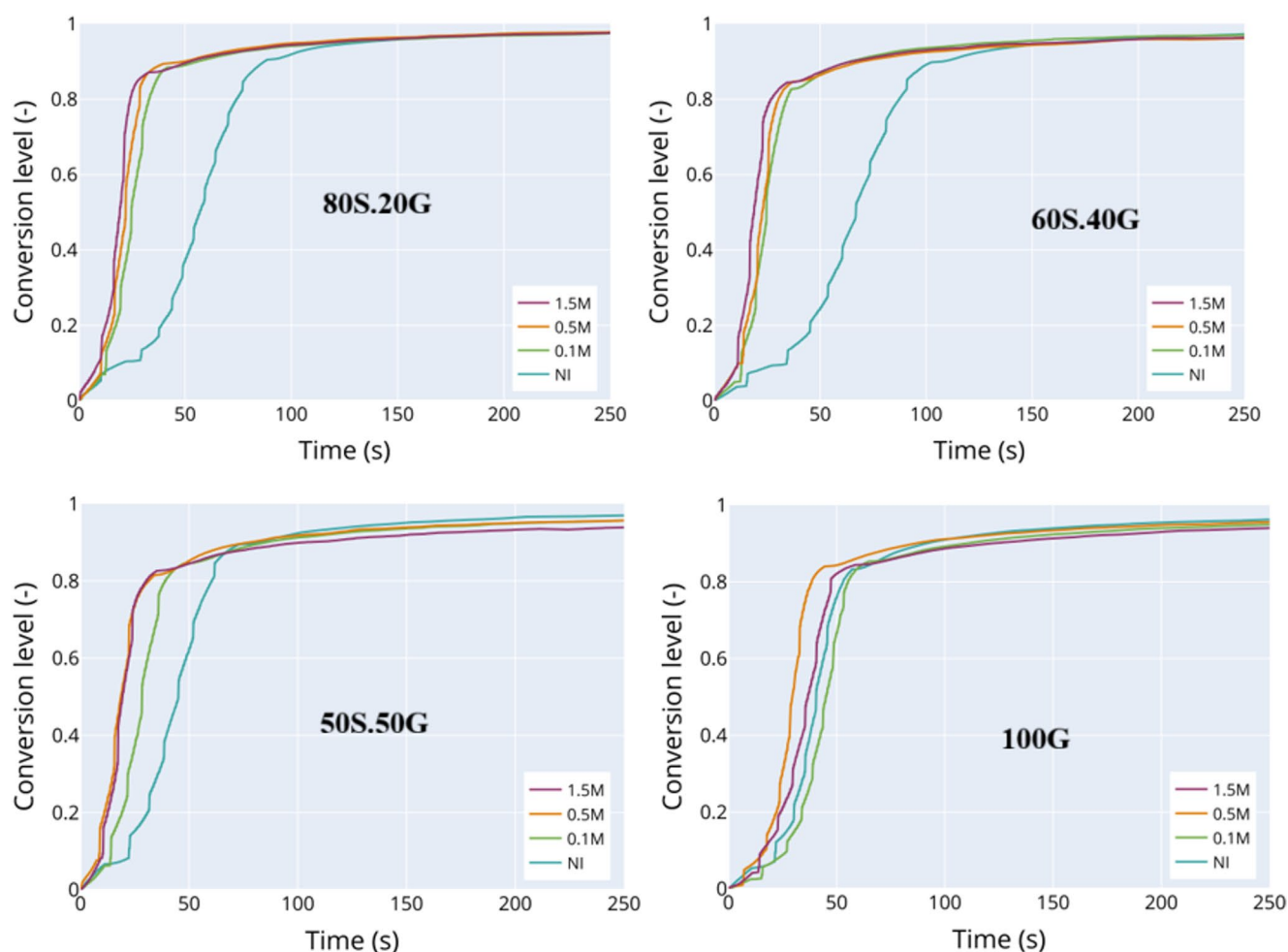


Fig. 4 The variation of the conversion of different samples as a function of time

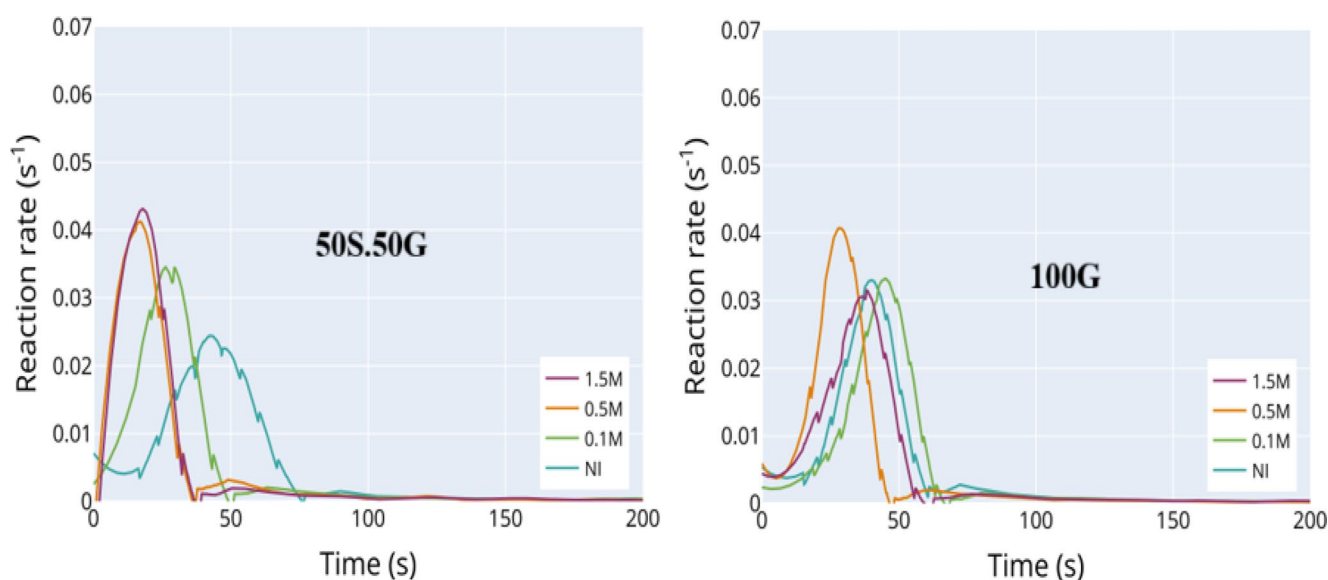


Fig. 5 Curves of conversion rate of 50 S.20G and 100G biomasses

Fig. 6 (SM), showing the pyrolysis of 100G and 80 S.20G without impregnation, where the 100G sample degrades faster than the 80 S.20G.

According to previous studies [37], pyrolysis curves generally exhibit a usual behavior during the degradation of dry lignocellulosic biomass which is defined by an initial rapid degradation and then a slow decomposition process: hemicellulose and cellulose decomposed rapidly in the first phase, followed by lignin, which decomposed slowly in the second stage. Mechanistically, the increase of the peak intensity corresponding to the biomass degradation after impregnation showed that potassium could accelerate the decomposition process of biomass, resulting in higher reactivity, known as the catalytic effect of potassium. Hence, potassium can increase the reactivity of biomass during pyrolysis and can lead to a decrease in the crystallinity of cellulose [40].

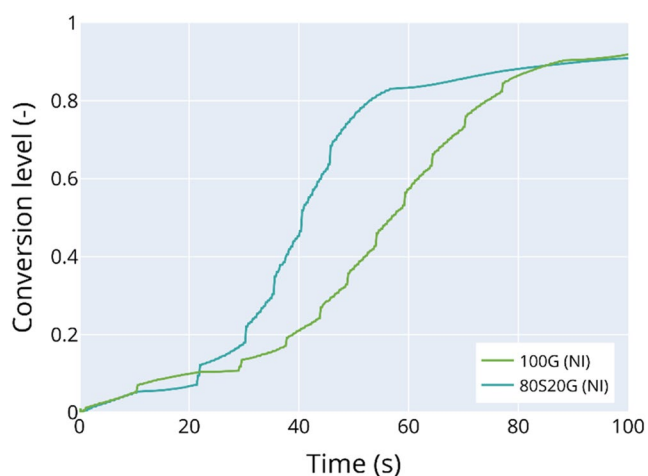


Fig. 6 Conversion versus time of 100G and 80S20G

However, beyond 0.1 M for the pine-rich samples, potassium's impact on degradation time remains almost constant. This suggests a saturating effect of the catalytic action of potassium on biomass degradation, implying that the biomass's surface could be covered by a monolayer of catalysts [41]. For the 100G and 50 S.50G samples, it is important to note that the slight variations observed at 1.5 M likely fall within the experimental margin of error rather than indicating a negative effect. This lack of significant catalytic improvement is highly correlated with the high inherent mineral load of the 100G biomass, which initially contains between 17.000 and 20.000 mg/kg of potassium (Table 1), thereby exceeding the critical saturation threshold for catalytic activity.

Furthermore, these kinetic variations are closely related to the percentage of hemicellulose, cellulose and lignin within each biomass type and the interactions between them. Lignin composition in pine-rich samples is of the order of 26% [42]. The distinct response of 50 S.50G and 100G samples with respect to high potassium concentrations can be explained by their high content of lignin, whose thermal degradation pathway is weakly influenced by alkali impregnation compared to cellulose and hemicellulose. While the 100G sample inherently degrades faster than the 80 S.20G sample, we hypothesize that this may be linked to the specific lignocellulosic composition, although other structural factors could also play a role since precise component measurements were not performed in this study.

The observed primary acceleration of mass loss is in agreement with Shoulaifar et al. [43], who showed that the effects of adding potassium to spruce wood on mass loss during torrefaction are significant, whether the potassium is added by doping or by impregnation with K_2CO_3 solutions.

Regarding the components' individual reactivity, our observations align with the conclusions of Shen et al. [44], who reported that potassium has a larger impact on cellulose and hemicelluloses degradation compared to lignin. In the same context, Hao et al. [15] affirmed not detecting any catalytic effect when adding KCl to isolated lignin fractions. Furthermore, several researches [45] found that the addition of potassium did not affect the temperature peak for each isolated polymer individually (hemicellulose, cellulose and lignin) since complex cross-reactions take place between these different constituents during the pyrolysis of a real blended biomass matrix.

3.3 Textural and Structure Characterization

Potassium impregnation significantly alters the chemical framework of the biomass and then the structure of chars generated after pyrolysis. As shown in Fig. 7 (supplemented materials (SM)), a macroscopic color variation of the 60 S.40G sample is observed following impregnation as a function of potassium concentration. This chromatic shift is mainly due to the ion (which induce the deprotonation of the acidic phenolic and carboxylic functional groups present within the lignin matrix [46]). This interaction causes the formation of phenoxide type of complexes and alters the light absorption properties of the biomass matrix [47].

3.4 SEM-EDS Analysis

Scanning Electron Microscopy (SEM) technique was used to observe the surface physical morphology of different char's samples. Selected SEM images of chars are presented in Figs. 8, 9, 10, 11, 12 and 13 and (SM). It can be highlighted that the impregnation process affects the texture of the biochar. As shown in Fig. 8 (SM), macropores in the non-impregnated chars are detectable. The morphology of the various impregnated char's sample is similar to that of the raw char, indicating that the substitution of monovalent or divalent cations did not generate any structural changes. Similarly, this applies for the surface of the char from the 80 S.20G sample that has been impregnated with potassium (Fig. 12 (SM)). Both two chars present irregular and porous surfaces.

On the other hand, by increasing the impregnation concentration up to 1.5 M, we observe in Fig. 10 (b) (SM) a flake structure appearing on the surface of the char. Aggregated deposits that plug the pores and are inhomogeneously distributed cover the char's surface, as indicated by orange arrows in Fig. 10 (c) (SM). In fact, the high concentration of potassium causes the aggregates to take on larger shapes.

By comparing the EDS mapping of potassium in the char (Fig. 8) with that of the char from the impregnated sample (0.5 M) (Fig. 9 (SM)), we notice that the dispersion of

potassium after impregnation is similar in both chars, which confirms the effectiveness of the impregnation process.

These results were also confirmed by EDX analyses (Figs. 14 and 15; Table 4) which were carried out on a particular area of the surface chars from 100G to 80 S.20G, both impregnated and non-impregnated, to show the distribution of potassium and other elements on the surface. The significant potassium peak in the curve (1.5 M) clearly illustrates the excess of this mineral accumulating on the surface of the char.

The specific surface area of biochar varies according to the type and content of mineral added. Mechanistically, since our selected pyrolysis temperature (400 °C) is significantly below the melting point of potassium carbonate (892 °C), these observed agglomerates are made up of potassium compounds, primarily potassium carbonate which remains either undecomposed or recrystallized. As a result, the extra potassium recrystallizes on the surface of the chars, leading to the formation of these larger aggregated shapes that plug the porous matrix.

This behavior is in line with previous findings, as a biochar with a low mineral content is generally characterized by a higher specific surface area than a biochar with a high ash content, as it was reported in the literature [48, 49]. Furthermore, our structural observations are in agreement with Bai et al. [50], affirming that the increased mineral content during char conversion leads to the formation of such surface agglomerates.

3.5 FTIR Analysis

Fourier transform infrared spectroscopy (FTIR) was performed to qualitatively investigate the effect of potassium impregnation on the chemical structure of different chars, and to evaluate any structure changes in the samples. Indeed, the sample's composition is identifiable through using relative intensities to analyze the spectra, which reveals various functional groups.

Figure 7 shows that the FTIR spectra of chars of non-impregnated samples exhibit roughly the same behavior and therefore the same structure, except for the band between 1550 cm^{-1} and 1600 cm^{-1} , which is attributed to the stretching vibrations of C=C of the aromatic skeletal characteristic of the complex structure of lignin [51, 52]. This specific band is more pronounced for 100G's char than 80S20G's char, which confirms that olive pomace char (100G char) is more enriched in lignin than pine sawdust char.

The effect of the impregnation with potassium can be easily observed in the FTIR spectra; certain bands are strongly amplified with larger potassium concentrations. FTIR spectra of impregnated chars show a high broad peak at around 3300 cm^{-1} ascribed to O-H stretching vibration in hydroxyl functional groups, whereas these vibration bands were not well detected in non-impregnated chars.

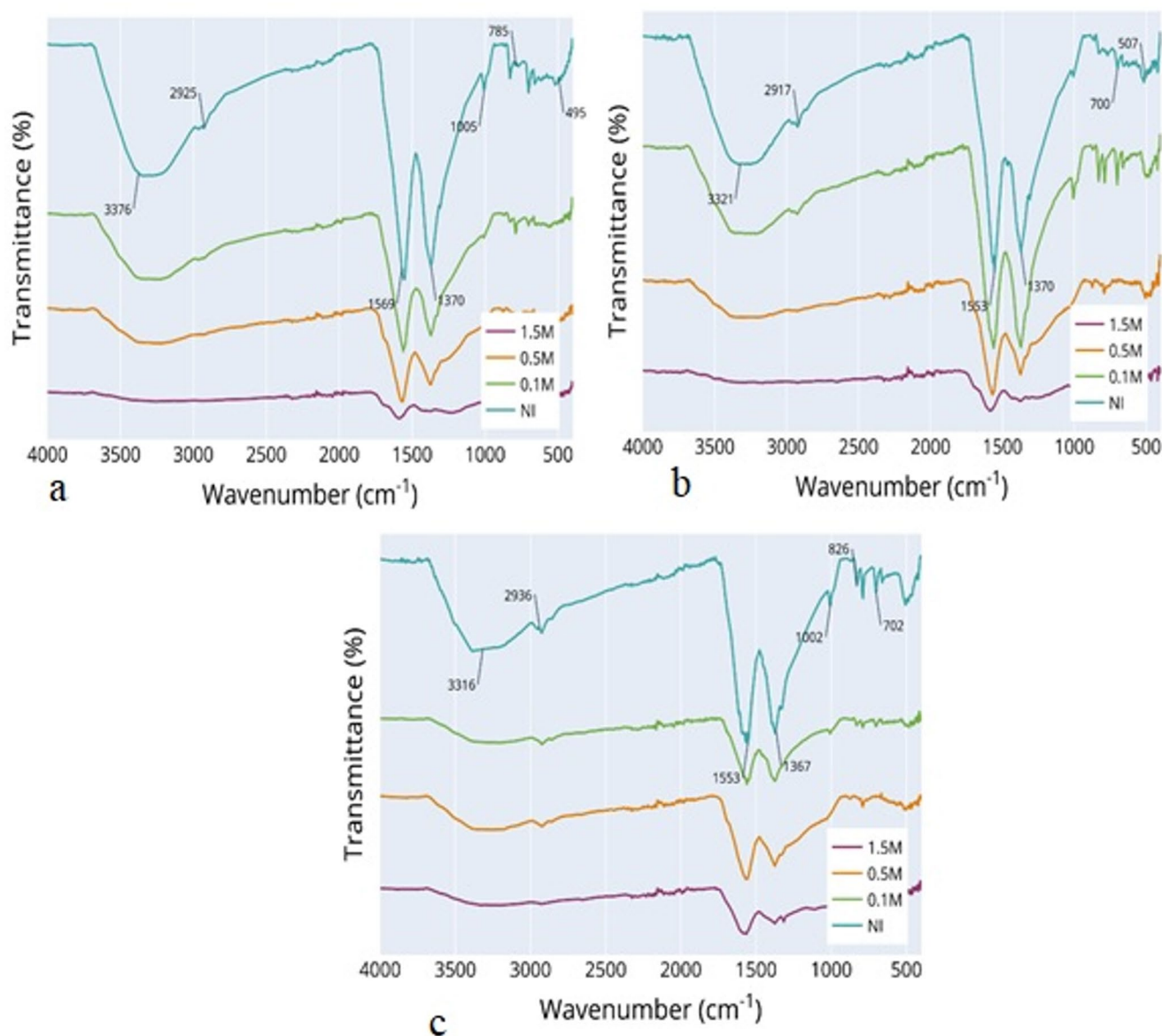


Fig. 7 FTIR spectra of different impregnated and non-impregnated chars ((a) 100G, (b) 50 S.50G, (c) 80 S.20G)

Small bands appear in the range of 2917 cm^{-1} and 2940 cm^{-1} corresponding to C-H stretching vibrations of aliphatic compounds (methyl and methylene groups). This is much more pronounced in the presence of potassium. Furthermore, the stretching signals in the range of 1000 cm^{-1} and 1040 cm^{-1} suggested the presence of C-O groups. The bands at 700 and 800 cm^{-1} are ascribed to stretching vibration of C-H in aromatic compounds [53]. These bands are not present in the FTIR spectra of non-impregnated sample chars.

For the treated samples, it is noticeable that by increasing the concentration of impregnated potassium, the peak situated in the range of 1550 cm^{-1} and 1600 cm^{-1} , associated with C=C stretching vibration, is considerably increased. Finally, a band around 1370 cm^{-1} appears for the treated samples.

The high broad peak observed at around 3300 cm^{-1} is due to the breaking of bonds by oxidation, which gives rise to -OH groups. These are characterized by the hydrogen bond and the formation of an aldehyde termination since the reducing medium does not allow the conversion to acids.

The fact that the aliphatic C-H stretching vibrations ($2917\text{--}2940\text{ cm}^{-1}$) become much more pronounced in the presence of potassium suggests that potassium can increase the stability and aromaticity of the char [54].

Regarding the sharp increase of the peak in the range of $1550\text{--}1600\text{ cm}^{-1}$ (C=C stretching), this behavior indicates that the polycondensation reaction of aromatic rings could be facilitated by potassium ions. This evolution also suggests the likely formation of carbonyl groups (notably

aldehydes), which can be explained by the reducing nature of the medium, limiting their conversion into acids [55]. Mechanistically, the novel appearance of the bands at 700 and 800 cm^{-1} is probably due to the intrinsic characteristics of K_2CO_3 , and the appearance of the band around 1370 cm^{-1} could be directly attributed to the transmission band of structural CO_3^{2-} anions retained within the solid matrix [56–58].

4 Conclusion

This paper demonstrates the significant influence of potassium impregnation on the pyrolysis of different biomass formulations, deeply affecting both product yields and char chemical structures.

A synthesis of the findings reveals that under previous pyrolysis conditions at a select temperature of 400 °C, macromolecular decomposition of cellulose and hemicellulose is almost complete, allowing for optimized solid retention. Initial reactivity is directly depend on the feedstock type, with the ash-rich olive mill residue (100G) showing faster early-stage degradation and enhanced baseline char yields compared to pine sawdust (100 S). After impregnation, potassium acts as a powerful catalyst that alters pyrolytic pathways, enhances polycondensation, and accelerates reaction kinetics. Quantitatively, this promotional effects leads to nearly a doubling of the final char yields for the blended biomass formulations (80 S.20G, 60 S.40G and 50 S.50G).

It can be highlighted that the catalytic efficiency is strictly concentration depend, pointing to a critical threshold concentration of potassium estimated to be around 18.000 mg/kg. Beyond this threshold, the catalytic effect on the reaction kinetics appears to level off, suggesting a progressive saturation of the available organic active sites. Furthermore, spectroscopic and morphological investigations (SEM, EDX and FTIR) suggest that an excess of potassium may induce structural alterations rather than further kinetic improvements, potentially altering the physical pore arrangement. Specifically, SEM-EDX analysis of the highly loaded 100G char exhibited that the excess unreacted or recrystallized potassium species tends to accumulate on the surface in the form of agglomerates.

Ultimately, this study enhances the understanding of potassium catalyzed thermal pathways. The multi-instrumental approach validates that potassium impregnation successfully activates supplementary chemical sites within the biochar framework and increases its aromatic stability. Such modifications in the structural and chemical properties are expected to play a crucial role in promoting reactivity during the subsequent steam gasification stage.

Figures 7 to 15 are provided in the Supplementary Materials File (SM).

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s42250-026-01742-5>.

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Data Availability Data will be provided on request.

Declarations

Ethical Approval Not applicable.

Informed Consent Not applicable.

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References

- Güleç F, Anburajan P, Umenweke GC, Umaru M, Williams O, Mortezaei Y et al (2024) Progress in lignocellulosic biomass valorization for biofuels and value-added chemical production in the EU: A focus on thermochemical conversion processes. *Biofuels Bioprod Biorefin* 18(3):755–781. <https://doi.org/10.1002/bbb.2513>
- Bioenergy IEA (2024) Literature review on social and economic sustainability of bioenergy systems. IEA Bioenergy
- Wafi T, Ben Othman A, Besbes M (2019) Qualitative and quantitative characterization of municipal solid waste and the unexploited potential of green energy in Tunisia. *Bioresources Bioprocess* 6:1–16. <https://doi.org/10.1186/s40643-019-0270-4>
- Danso-Boateng E, Achaw OW (2022) Bioenergy and biofuel production from biomass using thermochemical conversions technologies—A review. *AIMS Energy* 10(4):585–647. <https://doi.org/10.3934/energy.2022031>
- Jerzak W, Acha E, Li B (2024) Comprehensive review of biomass pyrolysis: conventional and advanced technologies, reactor designs, product compositions and yields, and techno-economic analysis. *Energies* 17(20):5082. <https://doi.org/10.3390/en17205082>
- Vinod A, Pulikkalparambil H, Jagadeesh P, Rangappa SM, Siengchin S (2023) Recent advancements in lignocellulose biomass-based carbon fiber: Synthesis, properties, and applications. *Heliyon* 9(3):e13842. <https://doi.org/10.1016/j.heliyon.2023.e13842>
- Wang W, Lemaire R, Bensakhria A, Luat D (2022) Analysis of the catalytic effects induced by alkali and alkaline earth metals (AAEMs) on the pyrolysis of beech wood and corncob. *Catalysts* 12(12):1505. <https://doi.org/10.3390/catal12121505>
- Guo F, Liu Y, Wang Y, Li X, Li T, Guo C (2016) Pyrolysis kinetics and behavior of potassium-impregnated pine wood in TGA and a fixed-bed reactor. *Energy Conv Manag* 130:184–191. <https://doi.org/10.1016/j.enconman.2016.10.022>

9. Jensen A, Dam-Johansen K, Wójtowicz MA, Serio MA (1998) TG-FTIR study of the influence of potassium chloride on wheat straw pyrolysis. *Energy Fuels* 12(5):929–938. <https://doi.org/10.1021/ef980053x>
10. Knudsen JN, Jensen PA, Dam-Johansen K (2004) Transformation and release to the gas phase of Cl, K, and S during combustion of annual biomass. *Energy Fuels* 18(5):1385–1399. <https://doi.org/10.1021/ef030180+>
11. Li P, Wang B, Hu J, Chen W, Fang S, Chang C, Pang S (2024) Characterization of the decomposition behaviors of catalytic pyrolysis of alkaline lignin with the addition of different concentrations of potassium. *J Energy Inst* 114:101618. <https://doi.org/10.1016/j.joei.2023.101618>
12. Bhunthong S, Aussawasathien D, Hrimchum K, Sriphalang SN (2018) Preparation and properties of activated carbon from palm shell by potassium hydroxide impregnation: Effects of processing parameters. *Chiang Mai J Sci* 45:462–473
13. Ray I, Mridha D, Roychowdhury T (2021) Waste derived amendments and their efficacy in mitigation of arsenic contamination in soil and soil–plant systems: A review. *Environ Technol Innov* 24:101976. <https://doi.org/10.1016/j.eti.2021.101976>
14. Fachini J, de Figueiredo CC, Vale AT, do (2022) Assessing potassium release in natural silica sand from novel K-enriched sewage sludge biochar fertilizers. *J Environ Manage* 314:115080. <https://doi.org/10.1016/j.jenvman.2022.115080>
15. Hao QL, Li BL, Liu L, Zhang ZB, Dou BJ, Wang WANG (2015) Effect of potassium on pyrolysis of rice husk and its components. *J Fuel Chem Technol* 43(1):34–41. [https://doi.org/10.1016/S1872-5813\(15\)30002-7](https://doi.org/10.1016/S1872-5813(15)30002-7)
16. Lajili M, Guizani C, Escudero-Sanz FJ, Jeguirim M (2018) Fast pyrolysis and steam gasification of pellets prepared from olive oil mill residues. *Energy* 150:61–68. <https://doi.org/10.1016/j.energy.2018.02.135>
17. Patwardhan PR, Satrio JA, Brown RC, Shanks BH (2010) Influence of inorganic salts on the primary pyrolysis products of cellulose. *Bioresour Technol* 101(12):4646–4655. <https://doi.org/10.1016/j.biortech.2010.01.112>
18. Radović LR, Walker PL Jr, Jenkins RG (1983) Importance of carbon active sites in the gasification of coal chars. *Fuel* 62:849–856. [https://doi.org/10.1016/0016-2361\(83\)90039-3](https://doi.org/10.1016/0016-2361(83)90039-3)
19. Kraiem N, Jeguirim M, Limousy L, Lajili M, Dorge S, Michelin L, Said R (2014) Impregnation of olive mill wastewater on dry biomasses: Impact on chemical properties and combustion performances. *Energy* 78:479–489. <https://doi.org/10.1016/j.energy.2014.10.061>
20. Chouchene A (2010) Etude expérimentale et théorique de procédés de valorisation de sous-produits oléicoles par voies thermique et physico-chimique (Doctoral dissertation, Université de Haute Alsace-Mulhouse)
21. Ribeiro TB, Oliveira AL, Costa C, Nunes J, Vicente AA, Pintado M (2020) Total and sustainable valorisation of olive pomace using a fractionation approach. *Appl Sci* 10(19):6785. <https://doi.org/10.3390/app10196785>
22. Guizani C, Escudero Sanz FJ, Salvador S (2013) The gasification reactivity of high-heating-rate chars in single and mixed atmospheres of H₂O and CO₂. *Fuel* 108:812–823. <https://doi.org/10.1016/j.fuel.2013.01.068>
23. Zhao M, Yang N, Yang B, Jiang Y, Zhang G (2007) Structural characterization of water-soluble polysaccharides from *Opuntia monacantha* cladodes in relation to their anti-glycated activities. *Food Chem* 105(4):1480–1486. <https://doi.org/10.1016/j.foodchem.2007.05.033>
24. De Jong W, Pirone A, Wojtowicz MA (2003) Pyrolysis of *Miscanthus giganteus* and wood pellets: TG-FTIR analysis and reaction kinetics. *Fuel* 82(9):1139–1147. [https://doi.org/10.1016/S0016-2361\(02\)00375-0](https://doi.org/10.1016/S0016-2361(02)00375-0)
25. Isahak WNRW, Hisham MW, Yarmo MA, Hin TYY (2012) A review on bio-oil production from biomass by using pyrolysis method. *Renew Sustain Energy Rev* 16(8):5910–5923. <https://doi.org/10.1016/j.rser.2012.06.018>
26. Collard FX, Blin J (2014) A review on pyrolysis of biomass constituents: Mechanisms and composition of the products obtained from the conversion of cellulose, hemicelluloses and lignin. *Renew Sustain Energy Rev* 38:594–608. <https://doi.org/10.1016/j.rser.2014.06.013>
27. Zhang L, Bao Z, Xia S, Lu Q, Walters KB (2018) Catalytic pyrolysis of biomass and polymer wastes. *Catalysts* 8(12):659
28. Yang H, Yan R, Chen H, Zheng C, Lee DH, Liang DT (2006) Influence of mineral matter on pyrolysis of palm oil wastes. *Combust Flame* 146(4):605–611. <https://doi.org/10.1016/j.combustflame.2006.04.005>
29. Nik-Azar M, Hajaligol MR, Sohrabi M, Dabir B (1997) Mineral matter effects in rapid pyrolysis of beech wood. *Fuel Process Technol* 51(1–2):7–17. [https://doi.org/10.1016/S0378-3820\(96\)01052-7](https://doi.org/10.1016/S0378-3820(96)01052-7)
30. Roy P, Dias G (2017) Prospects for pyrolysis technologies in the bioenergy sector: A review. *Renew Sustain Energy Rev* 77:59–69. <https://doi.org/10.1016/j.rser.2017.03.136>
31. Waters CL, Janupala RR, Mallinson RG, Lobban LL (2017) Staged thermal fractionation for segregation of lignin and cellulose pyrolysis products: An experimental study of residence time and temperature effects. *J Anal Appl Pyrol* 126:380–389. <https://doi.org/10.1016/j.jaap.2017.05.011>
32. Parihar MF, Kamil M, Goyal HB, Gupta AK, Bhatnagar AK (2007) An experimental study on pyrolysis of biomass. *Process Saf Environ Prot* 85(5):458–465. <https://doi.org/10.1205/psep07001>
33. Guizani C, Sanz FE, Salvador S (2013) The gasification reactivity of high-heating-rate chars in single and mixed atmospheres of H₂O and CO₂. *Fuel* 108:812–823. <https://doi.org/10.1016/j.fuel.2013.01.068>
34. Yang CY, Lu XS, Lin WG, Yang XM, Yao JZ (2006) TG-FTIR study on corn straw pyrolysis – Influence of minerals. *Chem Res Chin Univ* 22(4):524–532
35. Di Blasi C (2008) Modeling chemical and physical processes of wood and biomass pyrolysis. *Prog Energy Combust Sci* 34(1):47–90. <https://doi.org/10.1016/j.pecs.2006.12.001>
36. Nassar MM (1999) Thermal analysis kinetics of bagasse and rice straw. *Energy Sources* 21(1–2):131–137. <https://doi.org/10.1080/00908319950014365>
37. Nowakowski DJ, Jones JM (2008) Uncatalysed and potassium-catalysed pyrolysis of the cell-wall constituents of biomass and their model compounds. *J Anal Appl Pyrol* 83(1):12–25. <https://doi.org/10.1016/j.jaap.2008.05.008>
38. Perander M, DeMartini N, Brink A, Kramb J, Karlström O, Hemming J, Hupa M (2015) Catalytic effect of Ca and K on CO₂ gasification of spruce wood char. *Fuel* 150:464–472. <https://doi.org/10.1016/j.fuel.2015.02.026>
39. Struis RPWJ, Von Scala C, Stucki S, Prins R (2002) Gasification reactivity of charcoal with CO₂. Part I: Conversion and structural phenomena. *Chem Eng Sci* 57(17):3581–3592. [https://doi.org/10.1016/S0009-2509\(02\)00250-3](https://doi.org/10.1016/S0009-2509(02)00250-3)
40. Safar M, Lin BJ, Chen WH, Langauer D, Chang JS, Raclavská H et al (2019) Catalytic effects of potassium on biomass pyrolysis, combustion and torrefaction. *Appl Energy* 235:346–355. <https://doi.org/10.1016/j.apenergy.2018.10.062>
41. Struis RPWJ, Von Scala C, Stucki S, Prins R (2002) Gasification reactivity of charcoal with CO₂. Part II: Metal catalysis as a function of conversion. *Chem Eng Sci* 57(17):3593–3602. [https://doi.org/10.1016/S0009-2509\(02\)00251-5](https://doi.org/10.1016/S0009-2509(02)00251-5)

42. Rusanen A, Lappalainen K, Kärkkäinen J, Tuuttila T, Mikola M, Lassi U (2019) Selective hemicellulose hydrolysis of Scots pine sawdust. *Biomass Convers Biorefinery* 9:283–291. <https://doi.org/10.1007/s13399-017-0300-1>
43. Shoulaifar TK, DeMartini N, Karlström O, Hupa M (2016) Impact of organically bonded potassium on torrefaction: Part 1. *Experimental Fuel* 165:544–552. <https://doi.org/10.1016/j.fuel.2015.10.028>
44. Shen Y, Zhang N, Zhang S (2020) Catalytic pyrolysis of biomass with potassium compounds for co-production of high-quality bio-fuels and porous carbons. *Energy* 190:116431. <https://doi.org/10.1016/j.energy.2019.116431>
45. Shen Y, Zhang N, Zhang S (2020) Catalytic pyrolysis of biomass with potassium compounds for co-production of high-quality bio-fuels and porous carbons. *Energy* 190:116431. <https://doi.org/10.1016/j.energy.2019.116431>
46. Oriez V, Peydecastaing J, Pontalier PY (2020) Lignocellulosic biomass mild alkaline fractionation and resulting extract purification processes: conditions, yields, and purities. *Clean Technol* 2(1):91–115
47. D Evans P (2012) Weathering of wood and wood composites. *Handb wood Chem wood Compos* 2:151–216
48. Lehmann J et al (2015) Persistence of biochar in soil. *Biochar for environmental management*. Routledge, pp 235–282
49. Agblevor FA, Besler S (1996) Inorganic compounds in biomass feedstocks. 1. Effect on the quality of fast pyrolysis oils. *Energy Fuels* 10(2):293–298. <https://doi.org/10.1021/ef950195q>
50. Bai Y, Lv P, Yang X, Gao M, Zhu S, Yan L, Li F (2018) Gasification of coal char in H₂O/CO₂ atmospheres: Evolution of surface morphology and pore structure. *Fuel* 218:236–246. <https://doi.org/10.1016/j.fuel.2018.01.084>
51. Lin BJ, Colin B, Chen WH, Pétrissans A, Rousset P, Pétrissans M (2018) Thermal degradation and compositional changes of wood treated in a semi-industrial scale reactor in vacuum. *J Anal Appl Pyrol* 130:8–18. <https://doi.org/10.1016/j.jaap.2018.01.006>
52. Yang H, Yan R, Chen H, Lee DH, Zheng C (2007) Characteristics of hemicellulose, cellulose and lignin pyrolysis. *Fuel* 86(12–13):1781–1788. <https://doi.org/10.1016/j.fuel.2006.12.013>
53. Ke YH, Yang ET, Liu X, Liu CL, Dong WS (2013) Preparation of porous carbons from non-metallic fractions of waste printed circuit boards by chemical and physical activation. *Carbon* 60:563–573. <https://doi.org/10.1016/j.carbon.2013.04.031>
54. Ali R, Aslam Z, Shawabkeh RA, Asghar A, Hussein IA (2020) BET, FTIR, and RAMAN characterizations of activated carbon from waste oil fly ash. *Turk J Chem* 44(2):279–295. <https://doi.org/10.3906/kim-1910-37>
55. Guo F, Liu Y, Wang Y, Li X, Li T, Guo C (2016) Pyrolysis kinetics and behavior of potassium-impregnated pine wood in TGA and a fixed-bed reactor. *Energy Conv Manag* 130:184–191. <https://doi.org/10.1016/j.enconman.2016.10.022>
56. Molleti J, Yadav GD (2017) Potassium modified La–Mg mixed oxide as active and selective catalyst for mono-methylation of phenylacetone nitrile with dimethyl carbonate. *Mol Catal* 438:66–75. <https://doi.org/10.1016/j.mcat.2017.04.025>
57. Jindo K, Mizumoto H, Sawada Y, Sanchez-Monedero MA, Sonoki T (2014) Physical and chemical characterization of biochars derived from different agricultural residues. *Biogeosciences* 11(23):6613–6621. <https://doi.org/10.5194/bg-11-6613-2014>
58. Kan T, Strezov V, Evans TJ (2016) Lignocellulosic biomass pyrolysis: A review of product properties and effects of pyrolysis parameters. *Renew Sustain Energy Rev* 57:1126–1140. <https://doi.org/10.1016/j.rser.2015.12.185>

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